JSC370 2026: Final Project Report

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1. Introduction

Climate change is one of the major global challenges and is largely driven by carbon dioxide emissions from human activities. However, carbon dioxide emissions are not evenly distributed across countries. Notably, the differences in carbon dioxide emissions per-capita are closely related to variations in energy systems, economic structures, and levels of development in terms of GDP. Understanding these differences is important for designing effective climate and energy policies.

This project examines the following research question:

How have carbon dioxide emissions per capita evolved across countries from 1990 to 2023, and which economic, energy, and land-use factors explain and predict these differences?

To address this research question, this analysis uses data form 1990 to 2023 from the World Bank Open Data API, which provides a comprehensive set of development indicators covering economic, energy, and land-use factors. The dataset includes carbon dioxide emissions per capita (metric tons), GDP per capita (constant 2015 US dollars), renewable energy share, fossil-fuel electricity share, forest area as a percentage of land area, and the share of population living in urban areas. Furthermore, the year of observation is included as a key temporal indicator to account for global trends and technological shifts over time.

This analysis combines interpretable statistical models and machine learning methods to both explain and predict carbon dioxide emissions. Baseline models from the midterm, Linear Regression (OLS) and Generalized Additive Model (GAM), are estimated again and used as benchmarks. These are compared against tree-based machine learning models, such as Random Forest and XGBoost, which are designed to capture nonlinear relationships and interactions. Model performance is evaluated using a train, validation, and test framework and standard predictive metrics. In addition, variable importance measures are used to identify the most influential predictors and to compare findings across models.

2. Methods

The data for this project were obtained from World Bank Open Data platform through the World Bank Open Data API. The API provides access to the World Development Indicators (WDI) database, which contains standardized economic, energy, and land-use indicators for all countries and regions around the world. The analysis focuses on the period from 1990 to 2023, which ensures sufficient data availability for most countries. Several relevant indicators were retrieved from the API by a loop, including carbon dioxide emissions per capita, GDP per capita, renewable energy share, fossil-fuel electricity share, forest area as a percentage of total land area, and the share of the population living in urban areas. The API responses were in JSON format and converted into `pandas` DataFrames for further analysis by using the `requests` library in Python. All data were retrieved programmatically through the API to ensure reproducibility of the data acquisition process.

After retrieving the indicators from the API, several steps were taken to prepare the data for analysis. The raw API responses contain individual datasets for each indicator, so the first step was to organize these datasets into a consistent structure. The goal was to construct a dataset in which each row represents a country–year observation and each column represents a variable.

The datasets were merged using ISO3 country codes, i.e. three-letter country codes, and year identifiers. These variables provide consistent keys across World Bank indicators and allow observations from different datasets to be combined into a single table. The reason that not using the country name is because the country name might not be consistent across the different datasets although the data comes from the same source.

The separately retrieved indicator datasets were initially combined into a long-format dataset. If there were nested API fields, they were then flattened to extract the nested values into separate columns. As the World Bank dataset contains data for regions/aggregates as well as countries, such as Africa Eastern and Southern region, these regional aggregates were removed from the dataset to avoid misinterpretation. Therefore, it would be necessary to query the World Bank country metadata endpoint to collect all the country metadata in World Bank Open Data API. As the metadata contains a field called `region_name` for each country/region, when its value is "Aggregates", it means the row is a regional/group aggregate data rather than a country data. After collecting all the aggregate codes (name may also be fine, but codes are safer due to natural uniqueness), the dataset was filtered to retain only rows whose `countryiso3code` belongs to the valid country set. In effect, this step ensures that the final panel is composed only of individual countries, rather than regional summaries such as Africa Eastern and Southern or other World Bank aggregate groups. This is important because the project aims to compare country-level patterns, and aggregate units would distort those comparisons.

To facilitate the analysis, the long dataset was reshaped into a wide country-year dataset, where each indicator becomes a column instead of a value in a single column which stored all the indicator names/codes. This process produced a unified dataset containing economic, energy, land-use, and demographic indicators for each country and year. Data types were then standardized to ensure consistency across datasets. All the indicator variables were converted to float, as they are either percentages or metric tons or dollars, which are all numerical float values. Country identifiers and year variables were also checked to ensure consistent formatting across the merged datasets. Missing values were inspected for each variable before analysis. They were coded to `NaN` from all possible missing-value placeholders to ensure the consistency of later missing-value handling. Finally, several validation checks were performed to verify the integrity of the dataset. These checks included identifying duplicate country–year observations, formating of the country ISO3 codes, year range, and inspecting the range of indicator variables. These steps help ensure that the dataset is suitable for subsequent exploratory and statistical analysis.

For the main analysis, missing values were handled at the row level by dropping observations with missing values in the core analysis variables, while keeping all other available observations in the dataset until that stage. This approach keeps the missing-value treatment transparent and avoids introducing additional assumptions through imputation.

EDA was conducted to examine the structure of the dataset and to identify patterns related to the research question. First, descriptive statistics were computed for all numerical variables included in the dataset. Counts, means, medians, standard deviation, minimum & maximum values, and percentiles were calculated to summarize the distribution of each numerical variable and not including country codes and names. These measures provide an overview of the variables' distributions, which can help identify potential extreme values or outliers.

Visual exploration was then performed to examine patterns in the data. Several graphical methods were used to explore relationships between variables and to identify potential trends. Line plots were used to examine how carbon dioxide emissions per capita change over time. Histograms were used to inspect the distribution of individual variables and to identify skewness or potential outliers. Scatter plots were used to examine the relationship between carbon dioxide emissions per capita and explanatory variables. Box plots with the implementation of quantile binning were used to examine and confirm the relationship between carbon dioxide emissions per capita and explanatory variables that were identified in the scatter plots. So is the correlation matrix to examine the correlation between all the variables.

2.4 Modeling

TBD — chosen course-covered method (XGBoost / Random Forest / GAM / LDA). Feature engineering. Train/validation/test split. Hyperparameters. Metrics reported (R^2, RMSE, MAE, ROC/AUC, …).

3. Results

TBD — publication-ready, numbered, captioned tables and figures. Reference the website for interactive versions.

Figure 1. *Caption placeholder.*

Table 1. *Caption placeholder.*

4. Conclusions and Summary

TBD — interpret findings in the bigger picture, connect back to the research question, state limitations and possible extensions.